



Odd-odd continued fraction algorithm

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Abstract

By using a jump transformation associated to the Romik map, we define a new continued fraction algorithm called odd-odd continued fraction, whose principal convergents are rational numbers of odd denominators and odd numerators. Among others, it is proved that all the best approximating rationals of odd denominators and odd numerators of an irrational number are given by the principal convergents of the odd-odd continued fraction algorithm and vice versa.

Keywords Continued fractions · Diophantine approximation · Romik system

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1 Introduction

One main topic of Diophantine approximation studies the approximation of an irrational number by rational numbers. Given an irrational number x , we call a rational p/q a *best approximation* of x if

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$$|qx - p| < |bx - a| \text{ for any } \frac{a}{b} \neq \frac{p}{q} \text{ such that } 0 < b \leq q.$$

Here, and in the whole paper, by convention, when we write a rational number p/q , we always assume that $p \in \mathbb{Z}$, $q \in \mathbb{N}$ and p and q are coprime. The celebrated Lagrange Theorem (see [17, Chapter II] and [12, Section 6]) states that the best approximations of an irrational number x are the convergents, i.e., the finite truncations of the *regular continued fraction* (RCF) of x :

$$x = d_0 + \frac{1}{d_1 + \frac{1}{d_2 + \frac{1}{\ddots}}}, \quad (1)$$

where $d_0 \in \mathbb{Z}$ and $d_j \in \mathbb{N}$, for $j \geq 1$. More precisely, a rational p/q is a best approximation of an irrational x if and only if it is one of the convergents:

$$\frac{p_0^R}{q_0^R} := d_0, \quad \frac{p_1^R}{q_1^R} := d_0 + \frac{1}{d_1}, \quad \frac{p_2^R}{q_2^R} := d_0 + \frac{1}{d_1 + \frac{1}{d_2}}, \quad \dots$$

Let $\mathbb{H} = \{z \in \mathbb{C} : \text{Im}(z) > 0\}$ be the upper half-plane. The group $\text{SL}_2(\mathbb{R})$ acts on $\mathbb{H}$ as isometries defined by

$$g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} : z \mapsto g(z) = \frac{az + b}{cz + d}.$$

There is a close connection between the geodesics on the modular surface $\text{SL}_2(\mathbb{Z}) \backslash \mathbb{H}$ and the RCF algorithm (e.g. [22]). Especially, the orbit $\text{SL}_2(\mathbb{Z})(\infty) = \mathbb{Q}$ corresponds to a unique cusp of $\text{SL}_2(\mathbb{Z}) \backslash \mathbb{H}$. Let

$$\Theta = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \text{SL}_2(\mathbb{Z}) : \begin{pmatrix} a & b \\ c & d \end{pmatrix} \equiv \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \text{ or } \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \pmod{2} \right\}.$$

Then Θ is a subgroup of $\text{SL}_2(\mathbb{Z})$ of index 3 and the quotient space $\Theta \backslash \mathbb{H}$ is a hyperbolic surface with two cusps corresponding to the orbits $\Theta(\infty)$ and $\Theta(1)$ of ∞ and 1. Kraaikamp–Lopes [14] and Boca–Merriman [4] found that the geodesics on $\Theta \backslash \mathbb{H}$ is strongly related to the *even integer continued fraction* (EICF) introduced by Schweiger [19, 20], which is a continued fraction with even integers such that

$$b_0 + \frac{\eta_0}{b_1 + \frac{\eta_1}{b_2 + \ddots}}, \quad (2)$$

where $b_0 \in 2\mathbb{Z}$, $b_i \in 2\mathbb{N}$ for $i \geq 1$ and $\eta_i \in \{-1, +1\}$.

We classify rational numbers into two classes by the orbits $\Theta(\infty)$ and $\Theta(1)$. If $p/q \in \Theta(\infty)$, then p and q are of different parity. If $p/q \in \Theta(1)$, then p and q are both odd. We call a rational number in $\Theta(\infty)$ an ∞ -rational and a rational number in $\Theta(1)$ a 1-rational. The proportion of odd/even, even/odd and odd/odd in the RCF convergents was investigated by Moeckel [15]. Further, the asymptotic density of the RCF convergents whose denominators and numerators satisfying congruence equations was obtained by Jager-Liadet [10].

Short and Walker [23] defined a *best ∞ -rational approximation of x* by a rational $p/q \in \Theta(\infty)$ satisfying

$$|qx - p| < |bx - a| \text{ for any } \frac{a}{b} \in \Theta(\infty) \text{ apart from } \frac{p}{q} \text{ such that } 0 < b \leq q, \quad (3)$$

and showed that the best ∞ -rational approximations are convergents of EICF.

Our motivation of the paper is to study the best 1-rational approximations of an irrational number defined as follows.

Definition 1.1 For $x \in \mathbb{R} \setminus \mathbb{Q}$, $p/q \in \Theta(1)$ is a *best 1-rational approximation of x* if

$$|qx - p| < |bx - a| \text{ for any } \frac{a}{b} \in \Theta(1) \text{ apart from } \frac{p}{q} \text{ such that } 0 < b \leq q. \quad (4)$$

We introduce a new continued fraction, called the odd-odd continued fraction (OOCF, see Sect. 2) of the form

$$1 - \frac{1}{a_1 + \frac{\varepsilon_1}{2 - \frac{1}{a_2 + \frac{\varepsilon_2}{2 - \frac{1}{\ddots}}}}},$$

where $a_n \in \mathbb{N}$, and $\varepsilon_n \in \{1, -1\}$ for $a_n \geq 2$ and $\varepsilon_n = 1$ for $a_n = 1$. Our first main theorem is the following.

Theorem 1.2 A fraction p/q is a *best 1-rational approximation of an irrational number x* if and only if it is one of the principal convergents of the odd-odd continued fraction of x .

For RCF, Lagrange and Euler proved that an irrational number has eventually periodic RCF if and only if it is a quadratic irrational. (See [17, Chapter III-§1] and [12, Section 10]). For OOCF, we have the following second main theorem.

Theorem 1.3 An eventually periodic OOCF expansion converges to an ∞ -rational or a quadratic irrational. Moreover, a quadratic irrational has an eventually periodic OOCF expansion.

We also investigate the relation between the OOCF and the RCF. We show that for any real number x , the principal convergents of its OOCF are intermediate convergents of its RCF (Theorem 5.2). Further, we can convert RCF expansions into OOCF expansions (Theorem 5.3).

Our paper is organized as follows. In Sect. 2, we introduce the OOCF algorithm and give some basic properties of OOCFs. In Sect. 3 we study the principal convergents of OOCFs, and prove Theorem 1.3. Sect. 4 is devoted to the proof of Theorem 1.2. The relations between the OOCF expansions and the RCF expansions are described in the last section.

2 Odd-odd continued fraction algorithm

It is known that the *partial quotients* $d_j = d_j(x)$ of the RCF of an irrational number x as in (1) can be generated by the *Gauss map* $G : [0, 1] \rightarrow [0, 1]$ defined by

$$G(x) = \left\{ \frac{1}{x} \right\} \text{ for } x \in (0, 1] \text{ and } G(0) = 0,$$

where $\{\cdot\}$ is the fractional part. In fact, for an irrational x we have $d_j(x) = \lfloor 1/G^{j-1}(x) \rfloor$ for $j \geq 1$ with $\lfloor \cdot \rfloor$ being the integer part. Further, Gauss map is a jump transformation associated to the *Farey map* defined by

$$F(x) = \begin{cases} \frac{x}{1-x} & \text{if } 0 \leq x \leq \frac{1}{2}, \\ \frac{1-x}{x} & \text{if } \frac{1}{2} \leq x \leq 1. \end{cases}$$

In general, let $U : [0, 1] \rightarrow [0, 1]$ be a map and E be a subset of $[0, 1]$. The *first hitting time* of $x \in [0, 1]$ to E is defined by

$$n_E(x) := \min\{j \geq 0 : U^j(x) \in E\}.$$

A map $J : [0, 1] \rightarrow [0, 1]$ is called the *jump transformation associated to U with respect to E* (e.g. [21, Chapter 19]) if

$$J(x) = U^{n_E(x)+1}(x) \text{ for all } x \in [0, 1].$$

We can easily check that G is the jump transformation associated to F with respect to $E = \{0\} \cup (1/2, 1]$. In fact,

$$G(x) = F^{n_E(x)+1}(x) \text{ for } x \in [0, 1].$$

We also note that $n_E(x) + 1$ is exactly the first partial quotient a_1 of the RCF expansion of x .

Similar to the RCF, the partial quotients of EICF in (2) can be obtained by the EICF map $T_{\text{EICF}} : [0, 1] \rightarrow [0, 1]$ defined by

$$T_{\text{EICF}}(x) = \begin{cases} \frac{1}{x} - 2k & \text{if } \frac{1}{2k+1} \leq x \leq \frac{1}{2k}, \\ 2k - \frac{1}{x} & \text{if } \frac{1}{2k} \leq x \leq \frac{1}{2k-1}, \end{cases} \quad \text{for all } k \in \mathbb{N}, \quad \text{and } T_{\text{EICF}}(0) = 0.$$

The map T_{EICF} turns out to be a jump transformation of the following *Romik map*

$$R(x) = \begin{cases} \frac{x}{1-2x} & \text{if } 0 \leq x \leq \frac{1}{3}, \\ \frac{1}{x} - 2 & \text{if } \frac{1}{3} \leq x \leq \frac{1}{2}, \\ 2 - \frac{1}{x} & \text{if } \frac{1}{2} \leq x \leq 1, \end{cases} \quad (5)$$

introduced by Romik in [18]. In fact, letting $E_1 := \{0\} \cup [1/3, 1]$, we have

$$T_{\text{EICF}}(x) = R^{n_{E_1}(x)+1}(x) \quad \text{for } x \in [0, 1].$$

The Romik map was used to investigate an algorithm generating the Pythagorean triples by multiplying matrices [1–3, 7, 8]. Some number theoretical properties of the Romik map were recently shown in [5, 6]. Panti [16] studied the connection of the Romik map with the billiards in the hyperbolic plane.

Instead of E_1 , we choose $E_2 = [0, 1/2] \cup \{1\}$ and define $T_{\text{OOCF}} : [0, 1] \rightarrow [0, 1]$, called the *odd-odd continued fraction (OOCF) map*, by the jump transformation associated to the Romik map R with respect to E_2 , i.e.,

$$T_{\text{OOCF}}(x) = R^{n_{E_2}(x)+1}(x) \quad \text{for } x \in [0, 1].$$

By simple calculation, we have

$$T_{\text{OOCF}}(x) = \begin{cases} \frac{kx - (k-1)}{k - (k+1)x} & \text{if } \frac{k-1}{k} \leq x \leq \frac{2k-1}{2k+1}, \\ \frac{k - (k+1)x}{kx - (k-1)} & \text{if } \frac{2k-1}{2k+1} \leq x \leq \frac{k}{k+1}, \end{cases} \quad \text{for all } k \geq 1 \quad \text{and } T_{\text{OOCF}}(1) = 1. \quad (6)$$

The graph of T_{OOCF} is shown in Fig. 1.

Using T_{OOCF} , we can induce an OOCF expansion of $x \in [0, 1]$. For convenience, let $T := T_{\text{OOCF}}$. We partition $[0, 1]$ into the subintervals defined by

$$B(k+1, -1) := \left[\frac{k-1}{k}, \frac{2k-1}{2k+1} \right] \quad \text{and} \quad B(k, 1) := \left[\frac{2k-1}{2k+1}, \frac{k}{k+1} \right] \quad \text{for } k \in \mathbb{N}. \quad (7)$$

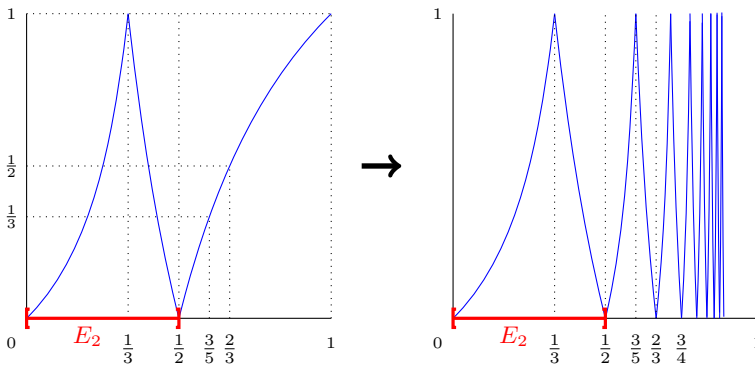


Fig. 1 The graph of R (left) and the graph of T_{OOCF} (right)

By (6), if $x \neq 1$, then

$$\frac{1}{1-x} = \begin{cases} (k+1) + \frac{-1}{2-(1-Tx)} & \text{if } x \in B(k+1, -1), \\ k + \frac{1}{2-(1-Tx)} & \text{if } x \in B(k, 1), \end{cases} \quad \text{for all } k \geq 1. \quad (8)$$

Thus, for all $x \notin \bigcup_{i=0}^{n-1} T^{-i}(\{1\})$

$$1-x = \frac{1}{a_1 + \frac{\varepsilon_1}{2 - \frac{1}{\ddots \frac{\varepsilon_n}{a_n + \frac{1}{2 - (1 - T^n x)}}}}}, \quad (9)$$

where

$$(a_n, \varepsilon_n) = \begin{cases} (k+1, -1), & \text{if } T^{n-1}(x) \in B(k+1, -1), \\ (k, 1), & \text{if } T^{n-1}(x) \in B(k, 1). \end{cases} \quad (10)$$

In particular, if $x = 0$, then $T(0) = 0$ and we have

$$1-x = \frac{1}{2 + \frac{-1}{2 - (1 - Tx)}}.$$

Thus, 0 has a unique infinite OOCF expansion: $0 = \llbracket (2, -1), (2, -1), \dots \rrbracket = \llbracket (2, -1)^\infty \rrbracket$. If $x = \frac{k}{k+1} \in B(k, 1) \cap B(k+2, -1)$ for some $k \geq 1$. Then $T(x) = 0$ and we have both choices in (8):

$$1 - x = \frac{1}{k+1} = \frac{1}{k+2 + \frac{-1}{2 - (1 - Tx)}} = \frac{1}{k + \frac{1}{2 - (1 - Tx)}}.$$

Hence, a rational number $x = \frac{k}{k+1}$ has two infinite OOCF expansions: $x = \llbracket (k+2, -1), (2, -1)^\infty \rrbracket$ and $x = \llbracket (k, 1), (2, -1)^\infty \rrbracket$. Furthermore, for any $x \in T^{-n}(\{0\})$ with some $n \geq 2$, we can apply the iteration (8) $n-1$ times, and then write $1 - T^{n-1}(x)$ in two different ways. Therefore, any $x \in \bigcup_{n \geq 1} T^{-n}(\{0\})$ has two infinite OOCF expansions ending with $(2, -1)^\infty$.

Let $x = \frac{2k-1}{2k+1} \in B(k+1, -1) \cap B(k, 1)$ for some $k \geq 1$. Then $T(x) = 1$ and

$$\begin{aligned} 1 - x &= \frac{1}{(k+1) + \frac{-1}{2 - (1 - Tx)}} = \frac{1}{(k+1) + \frac{-1}{2}}, \quad \text{or} \\ 1 - x &= \frac{1}{k + \frac{1}{2 - (1 - Tx)}} = \frac{1}{k + \frac{1}{2}}. \end{aligned}$$

Thus $x = \frac{2k-1}{2k+1}$ has two finite OOCF expansions: $x = \llbracket (k+1, -1) \rrbracket$ and $x = \llbracket (k, 1) \rrbracket$. Furthermore, for any $x \in T^{-n}(\{1\})$ with some $n \geq 1$ and $x \neq 1$, we can apply the iteration (8) $n-1$ times, and then write $1 - T^{n-1}(x)$ in two different ways. Therefore, any $x \in \bigcup_{n \geq 0} T^{-n}(\{1\}) \setminus \{1\}$ has two finite OOCF expansions which differ at the last partial quotient.

Note that for a rational m/n , the denominator of $T_{\text{OOCF}}(m/n)$ is strictly less than n . Note also that T_{OOCF} sends a 1-rational to a 1-rational and an ∞ -rational to an ∞ -rational. Thus if m/n is a 1-rational, then $T_{\text{OOCF}}^N(m/n) = 1$ for some $N \geq 1$, while if m/n is a non-zero ∞ -rational, then $T_{\text{OOCF}}^N(m/n) = 0$ for some $N \geq 1$. Hence any 1-rational and ∞ -rational belongs to $T_{\text{OOCF}}^{-N}(\{1\})$ and $T_{\text{OOCF}}^{-N}(\{0\})$ for some $N \geq 1$ respectively. Finally, we note that

$$\bigcup_{n=1}^{\infty} T_{\text{OOCF}}^{-n}(\{0, 1\}) = \mathbb{Q}.$$

Then for any $x \in [0, 1] \setminus \mathbb{Q}$, we can iterate (9) infinitely and uniquely to get its OOCF expansion:

$$x = 1 - \frac{1}{a_1 + \frac{\varepsilon_1}{2 - \frac{1}{a_2 + \frac{\varepsilon_2}{2 - \frac{1}{\ddots}}}}}.$$

where $a_n \in \mathbb{N}$ and $\varepsilon_n \in \{1, -1\}$ for $a_n \geq 2$ and $\varepsilon_n = 1$ for $a_n = 1$.

We denote the OOCF expansion of x by

$$x = \llbracket (a_1, \varepsilon_1), (a_2, \varepsilon_2), \dots, (a_n, \varepsilon_n), \dots \rrbracket,$$

and call (a_n, ε_n) the n -th partial quotients of x in its OOCF expansion. By the above discussion, we have the following proposition.

Proposition 2.1 *The following properties hold.*

- (1) Any non-zero ∞ -rational has exactly two infinite OOCF expansions ending with $(2, -1)^\infty$.
- (2) Each 1-rational has exactly two finite OOCF expansions which differ only in the last partial quotient.
- (3) Every irrational has a unique infinite OOCF expansion.

To end this section, we remark that the two maps T_{OOCF} and T_{EICF} are conjugate. Define $f : [0, 1] \rightarrow [0, 1]$ by $f(x) := \frac{1-x}{1+x}$. Then the map T_{OOCF} is conjugate to T_{EICF} via f , i.e.,

$$f \circ T_{\text{OOCF}} \circ f^{-1} = T_{\text{EICF}}.$$

Schweiger [19] proved that T_{EICF} admits an ergodic absolutely continuous invariant measure: $d\mu := \frac{dx}{1-x^2}$. Thus, the measure $f_*^{-1}\mu$ is an ergodic absolutely continuous invariant measure with respect to T_{OOCF} . Denote by $y = f(x)$. We have

$$\frac{dx}{1-x^2} = \frac{(1+x)^2 dy}{2(1-x^2)} = \frac{(1+x)dy}{2(1-x)} = \frac{dy}{2y}.$$

Hence, we have the following conclusion.

Proposition 2.2 *The map $T_{\text{OOCF}} : [0, 1] \rightarrow [0, 1]$ preserves an infinite ergodic absolutely continuous invariant measure $\frac{1}{x}dx$.*

We also remark that the infinity of the absolutely continuous invariant measure comes from the fact that the map T_{OOCF} has 0 as an indifferent fixed point.

3 Convergents of the odd-odd continued fraction algorithm

For OOCF, we have three types of convergents by truncating the OOCF in three different places. We will investigate the basic properties of such convergents of OOCF.

Let $x \in (0, 1)$ be a real number such that

$$x = \llbracket (a_1, \varepsilon_1), (a_2, \varepsilon_2), \dots, (a_n, \varepsilon_n), \dots \rrbracket.$$

For $n \geq 1$, the n -th principal convergent of OOCF is defined by

$$\frac{p_n}{q_n} = \llbracket (a_1, \varepsilon_1), (a_2, \varepsilon_2), \dots, (a_n, \varepsilon_n) \rrbracket = 1 - \frac{1}{a_1 + \frac{\varepsilon_1}{2 - \frac{1}{\ddots \frac{\varepsilon_{n-1}}{2 - \frac{1}{a_n + \frac{\varepsilon_n}{2}}}}}}.$$

We denote

$$\frac{p'_n}{q'_n} := 1 - \frac{1}{a_1 + \frac{\varepsilon_1}{2 - \frac{1}{\ddots \frac{\varepsilon_{n-1}}{2 - \frac{1}{a_n}}}}} \quad \text{and} \quad \frac{p''_n}{q''_n} := 1 - \frac{1}{a_1 + \frac{\varepsilon_1}{2 - \frac{1}{\ddots \frac{\varepsilon_{n-1}}{2 - \frac{1}{a_n + \varepsilon_n}}}}},$$

and call them the n -th sub-convergent and n -th pseudo-convergent, respectively.

To study the convergents of a continued fraction, we have the following general lemma proved by induction (see [13, p. 3] for details).

Lemma 3.1 *Consider a general infinite continued fraction and its truncated continued fraction of the form*

$$g_0 + \frac{e_0}{g_1 + \frac{e_1}{g_2 + \frac{e_2}{g_3 + \ddots}}} \quad \text{and} \quad \frac{r_n}{s_n} := g_0 + \frac{e_0}{g_1 + \frac{e_1}{g_2 + \frac{e_2}{\ddots + \frac{e_{n-1}}{g_n}}}},$$

where $g_n \in \mathbb{Z}$ and $|e_n| = 1$. Then the following matrix relation holds:

$$\begin{pmatrix} r_n & e_n r_{n-1} \\ s_n & e_n s_{n-1} \end{pmatrix} = \begin{pmatrix} g_0 & e_0 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} g_1 & e_1 \\ 1 & 0 \end{pmatrix} \cdots \begin{pmatrix} g_n & e_n \\ 1 & 0 \end{pmatrix}.$$

Consequently, we have the following recursive formulas:

$$\begin{cases} r_n = g_n r_{n-1} + e_{n-1} r_{n-2}, \\ s_n = g_n s_{n-1} + e_{n-1} s_{n-2}, \end{cases}$$

where $r_{-1} = 1$, $s_{-1} = 0$, $r_0 = g_0$ and $s_0 = 1$.

Denote the inverse of $T_{\text{OOCF}}|_{B(a_n, \varepsilon_n)}$ by $f_{(a_n, \varepsilon_n)}$. Then by (8), we have

$$f_{(a_n, \varepsilon_n)}(t) = 1 - \frac{1}{a_n + \frac{\varepsilon_n}{1+t}}. \quad (11)$$

The map $f_{(a_n, \varepsilon_n)}$ corresponds to a linear fractional map on the upper half-plane $\mathbb{H}$ given by the matrix

$$A_{(a_n, \varepsilon_n)} := \begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} a_n & \varepsilon_n \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} a_n - 1 & a_n + \varepsilon_n - 1 \\ a_n & a_n + \varepsilon_n \end{pmatrix} \in \Theta \cup \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \Theta. \quad (12)$$

By Lemma 3.1, we have

$$\begin{aligned} A_{(a_1, \varepsilon_1)} A_{(a_2, \varepsilon_2)} \cdots A_{(a_n, \varepsilon_n)} \begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix} &= \begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} a_1 & \varepsilon_1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 2 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} a_2 & \varepsilon_2 \\ 1 & 0 \end{pmatrix} \\ &\cdots \begin{pmatrix} 2 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} a_n & \varepsilon_n \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 2 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} p_n & -p'_n \\ q_n & -q'_n \end{pmatrix} \end{aligned} \quad (13)$$

and

$$\begin{aligned} A_{(a_1, \varepsilon_1)} A_{(a_2, \varepsilon_2)} \cdots A_{(a_n, \varepsilon_n)} \begin{pmatrix} 0 \\ 1 \end{pmatrix} \\ = \begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} a_1 & \varepsilon_1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 2 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} a_2 & \varepsilon_2 \\ 1 & 0 \end{pmatrix} \cdots \begin{pmatrix} 2 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} a_n & \varepsilon_n \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} p''_n \\ q''_n \end{pmatrix} \end{aligned} \quad (14)$$

These mean that under the linear fractional map of the matrix $A_{(a_1, \varepsilon_1)} A_{(a_2, \varepsilon_2)} \cdots A_{(a_n, \varepsilon_n)}$, the images of 1, ∞ and 0 are p_n/q_n , p'_n/q'_n and p''_n/q''_n , respectively.

Since $A_{(a_1, \varepsilon_1)} A_{(a_2, \varepsilon_2)} \cdots A_{(a_n, \varepsilon_n)}$ is contained in Θ or $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \Theta$, we deduce the following proposition.

Proposition 3.2 *We have*

$$\frac{p_n}{q_n} \in \Theta(1) \quad \text{and} \quad \frac{p'_n}{q'_n}, \frac{p''_n}{q''_n} \in \Theta(\infty).$$

We remark that the name of odd-odd continued fraction comes from the fact that the principal convergents p_n/q_n are 1-rationals, i.e., of odd denominators and odd numerators. We also remark that by Proposition 3.2, any finite OOCF is a 1-rational.

By (13) and (14), we have the following recursive relations of the three types of convergents.

Lemma 3.3 *Let $p'_0 = 1$, $q'_0 = 0$, $p_0 = 1$ and $q_0 = 1$. We have the following recursive formulas:*

$$\begin{cases} p'_n = a_n p_{n-1} - p'_{n-1}, \\ q'_n = a_n q_{n-1} - q'_{n-1}, \end{cases} \quad \begin{cases} p''_n = p'_n + \varepsilon_n p_{n-1}, \\ q''_n = q'_n + \varepsilon_n q_{n-1} \end{cases} \quad \text{and} \\ \begin{cases} p_n = 2p'_n + \varepsilon_n p_{n-1}, \\ q_n = 2q'_n + \varepsilon_n q_{n-1} \end{cases} \quad \text{for } n \geq 1. \quad (15)$$

Further,

$$\begin{cases} p_n = p'_n + p''_n, \\ q_n = q'_n + q''_n \end{cases} \quad \text{and} \quad \begin{cases} p_{n-1} = \varepsilon_n (p''_n - p'_n), \\ q_{n-1} = \varepsilon_n (q''_n - q'_n) \end{cases} \quad \text{for } n \geq 1. \quad (16)$$

Moreover, letting $p_{-1} = -1$, $q_{-1} = 1$ and $\varepsilon_0 = 1$, we have the recursive formulas for the principal convergents

$$\begin{cases} p_n = (2a_n + \varepsilon_n - 1)p_{n-1} + \varepsilon_{n-1}p_{n-2}, \\ q_n = (2a_n + \varepsilon_n - 1)q_{n-1} + \varepsilon_{n-1}q_{n-2} \end{cases} \quad \text{for } n \geq 1. \quad (17)$$

Note that $2a_n + \varepsilon_n - 1 \geq 2$ for all $n \in \mathbb{N}$. Since $q_1 \geq 2q_0 + \varepsilon_0 q_{-1} = 3 > q_0$, by (17), we have

$$q_{n+1} > q_n, \quad \text{for all } n \in \mathbb{N}. \quad (18)$$

By (13) and the second assertion of (16), we have

$$A_{(a_1, \varepsilon_1)} A_{(a_2, \varepsilon_2)} \cdots A_{(a_n, \varepsilon_n)} \begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} \varepsilon_n p_{n-1} & p_n \\ \varepsilon_n q_{n-1} & q_n \end{pmatrix} \quad \text{for all } n \in \mathbb{N}.$$

Thus, by noting $\det(A_{(a_i, \varepsilon_i)}) = \varepsilon_i$, we have

$$p_{n-1}q_n - p_n q_{n-1} = -2\varepsilon_1 \cdots \varepsilon_{n-1} \quad \text{for all } n \in \mathbb{N}. \quad (19)$$

Denote by

$$\zeta_n := T_{\text{OOCF}}^{n-1}(x) = 1 - \frac{1}{a_n + \frac{\varepsilon_n}{2 - \frac{1}{a_{n+1} + \frac{\varepsilon_{n+1}}{\ddots}}}}.$$

We can show inductively that

$$x = \frac{p_n'' + p_n' \zeta_{n+1}}{q_n'' + q_n' \zeta_{n+1}} \quad \text{for all } n \in \mathbb{N}. \quad (20)$$

By (16), we have

$$\begin{cases} p_n'' = \frac{1}{2}(p_n + \varepsilon_n p_{n-1}), \\ p_n' = \frac{1}{2}(p_n - \varepsilon_n p_{n-1}), \end{cases} \quad \text{and} \quad \begin{cases} q_n'' = \frac{1}{2}(q_n + \varepsilon_n q_{n-1}), \\ q_n' = \frac{1}{2}(q_n - \varepsilon_n q_{n-1}), \end{cases} \quad \text{for all } n \in \mathbb{N}.$$

Thus,

$$x = \frac{p_n(1 + \zeta_{n+1}) + \varepsilon_n p_{n-1}(1 - \zeta_{n+1})}{q_n(1 + \zeta_{n+1}) + \varepsilon_n q_{n-1}(1 - \zeta_{n+1})}. \quad (21)$$

The following theorem gives the convergence of our OOCF.

Theorem 3.4 *For all $x \in [0, 1]$, the OOCF expansion of x converges to x .*

Proof Let $\xi_n = \frac{1-\zeta_n}{1+\zeta_n}$. By (21) and (19), we have

$$x - \frac{p_n}{q_n} = \frac{\xi_{n+1} \varepsilon_n (p_{n-1} q_n - q_{n-1} p_n)}{q_n (q_n + \varepsilon_n q_{n-1} \xi_{n+1})} = \frac{-2\xi_{n+1} \varepsilon_1 \cdots \varepsilon_n}{q_n (q_n + \varepsilon_n q_{n-1} \xi_{n+1})}. \quad (22)$$

Since $|\varepsilon_n| = 1$ and $|\xi_{n+1}| \leq 1$, by (18), we have

$$\left| x - \frac{p_n}{q_n} \right| < \frac{2}{q_n} \quad \text{for all } n \in \mathbb{N}.$$

Again by (18), $q_n \rightarrow \infty$ as $n \rightarrow \infty$, which concludes $p_n/q_n \rightarrow x$ as $n \rightarrow \infty$. $\square$

Lemma 3.5 *For all $x \in (0, 1)$, x is between p_n/q_n and p_n''/q_n'' .*

Proof The equations (13) and (14) imply that $p_n' q_n'' - p_n'' q_n' = \varepsilon_1 \cdots \varepsilon_n$. Thus, by (20),

$$x - \frac{p_n''}{q_n''} = \frac{\zeta_{n+1} (p_n' q_n'' - p_n'' q_n')}{q_n'' (q_n'' + q_n' \zeta_{n+1})} = \frac{\zeta_{n+1} \varepsilon_1 \cdots \varepsilon_n}{q_n'' (q_n'' + q_n' \zeta_{n+1})},$$

which means that $x - p_n/q_n$ and $x - p_n''/q_n''$ has opposite signs by comparing with (22). $\square$

With the above preparations, we can now show the following lemma.

Lemma 3.6 *The following statements hold.*

- (1) *The n -th principal convergent p_n/q_n is between p_n'/q_n' and p_n''/q_n'' .*
- (2) *The $(n-1)$ -th principal convergent p_{n-1}/q_{n-1} is not between p_n'/q_n' and p_n''/q_n'' .*

(3) The three distinct convergents p_n/q_n , p'_n/q'_n and p''_n/q''_n are in the half closed interval I_{n-1} of endpoints p_{n-1}/q_{n-1} and p''_{n-1}/q''_{n-1} which contains p''_{n-1}/q''_{n-1} but does not contain p_{n-1}/q_{n-1} .

Proof The first two assertions follow from (16) and the fact that for two rationals a/b and c/d such that $bd > 0$, if $a/b \leq c/d$, then

$$\frac{a}{b} \leq \frac{a+c}{b+d} \leq \frac{c}{d}.$$

For (3), by Lemma 3.3, we have

$$\begin{cases} p'_n = (a_n - 1)p_{n-1} + p''_{n-1}, \\ q'_n = (a_n - 1)q_{n-1} + q''_{n-1}, \end{cases} \quad \text{and} \quad \begin{cases} p''_n = (a_n - 1 + \varepsilon_n)p_{n-1} + p''_{n-1}, \\ q''_n = (a_n - 1 + \varepsilon_n)q_{n-1} + q''_{n-1}. \end{cases}$$

Since $a_n - 1 \geq 0$ and $a_n + \varepsilon_n - 1 \geq 0$, both p'_n/q'_n and p''_n/q''_n are in I_{n-1} . By the first assertion, p_n/q_n is also in I_{n-1} . $\square$

Now, we are ready to prove Theorem 1.3.

Proof of Theorem 1.3 If x has an eventually periodic OOCF, then there exist distinct positive integers i and j such that $T^i(x) = T^j(x)$. Since T^i and T^j are linear fractional maps, $T^i(x)$ is either 0 or a quadratic irrational. In the former case x is an ∞ -rational, while in the latter case x is a quadratic irrational.

For the second assertion, let x be a quadratic irrational between 0 and 1 such that $\alpha_1 x^2 + \beta_1 x + \gamma_1 = 0$ where α_1 , β_1 and γ_1 are coprime integers. By (20), we have for all $i \geq 1$,

$$\alpha_1(p''_i + p'_i \zeta_{i+1})^2 + \beta_1(p''_i + p'_i \zeta_{i+1})(q''_i + q'_i \zeta_{i+1}) + \gamma_1(q''_i + q'_i \zeta_{i+1})^2 = 0.$$

For $i \geq 1$, let

$$\begin{aligned} \alpha_{i+1} &= \alpha_1(p'_i)^2 + \beta_1 p'_i q'_i + \gamma_1 (q'_i)^2, \\ \beta_{i+1} &= 2\alpha_1 p''_i p'_i + \beta_1(p''_i q'_i + p'_i q''_i) + 2\gamma_1 q''_i q'_i, \\ \gamma_{i+1} &= \alpha_1(p''_i)^2 + \beta_1 p''_i q''_i + \gamma_1 (q''_i)^2. \end{aligned} \quad (23)$$

Then

$$\alpha_{i+1} \zeta_{i+1}^2 + \beta_{i+1} \zeta_{i+1} + \gamma_{i+1} = 0. \quad (24)$$

Since $|q'_i p''_i - q''_i p'_i| = 1$, we can check that

$$\beta_{i+1}^2 - 4\alpha_{i+1}\gamma_{i+1} = \beta_1^2 - 4\alpha_1\gamma_1. \quad (25)$$

On the other hand, we also have $|(x - p'_i/q'_i) + (p''_i/q''_i - x)| = 1/q'_i q''_i$. Thus, by Lemma 3.5 and the fact that p_i/q_i is between p'_i/q'_i and p''_i/q''_i , we deduce that $x - p'_i/q'_i$ and

$x - p_i''/q_i''$ have opposite signs. Thus, $|q_i'x - p_i'| < 1/q_i''$ and $|p_i'' - q_i''x| < 1/q_i'$. Hence, there are $|\delta| < 1$ and $|\lambda| < 1$ such that

$$p_i' = q_i'x + \delta/q_i'' \quad \text{and} \quad p_i'' = q_i''x + \lambda/q_i'. \quad (26)$$

By plugging (26) in (23), we derive the following expressions

$$\alpha_{i+1} = \delta \left(\frac{q_i'}{q_i''} (2\alpha_1x + \beta_1) + \alpha_1 \frac{\delta}{(q_i'')^2} \right), \quad (27)$$

$$\beta_{i+1} = (2\alpha_1x + \beta_1)(\delta + \lambda) + 2\alpha_1 \frac{\delta\lambda}{q_i'q_i''} \quad \text{and} \quad (28)$$

$$\gamma_{i+1} = \lambda \left(\frac{q_i''}{q_i'} (2\alpha_1x + \beta_1) + \alpha_1 \frac{\lambda}{(q_i')^2} \right). \quad (29)$$

By (28), we have $|\beta_{i+1}| \leq 2(|2\alpha_1| + |\beta_1|) + |2\alpha_1|$, thus the coefficient β_{i+1} is bounded. If $q_i'' \geq q_i'$, then by (27), we have $|\alpha_{i+1}| < 2|\alpha_1| + |\beta_1| + |\alpha_1|$. Thus, α_{i+1} is bounded. Further, by (25), γ_{i+1} is bounded. Similarly, if $q_i'' < q_i'$, then by (29), γ_{i+1} is bounded since $|\gamma_{i+1}| < 2|\alpha_1| + |\beta_1| + |\alpha_1|$. Moreover, by (25), α_{i+1} is also bounded. Thus in all cases, the coefficients of the equation (24) are all bounded. Therefore, $\{\zeta_i\}_{i \in \mathbb{N}}$ has only finitely many values which means that $\zeta_n = \zeta_m$ for some m and n . Therefore, the OOCF expansion of x is eventually periodic. $\square$

Remark 3.7 From the proof of Proposition 2.1-(1), we see that if x is an ∞ -rational, then its OOCF ends with $(2, -1)^\infty$ and thus there exists $n_0 \geq 0$ such that $\zeta_{n+1} = 0$ for all $n \geq n_0$. Hence, by (20), we have $x = p_n''/q_n''$ for all $n \geq n_0$.

At the end of this section, let us discuss the relation between the OOCF convergents of a number x and the EICF convergents of $1 - x$.

We denote the EICF expansion in (2) by a sequence in a double angle bracket:

$$\langle\langle (b_1, \eta_1), (b_2, \eta_2), \dots, (b_n, \eta_n), \dots \rangle\rangle = \langle\langle (b_n, \eta_n)_{n \in \mathbb{N}} \rangle\rangle.$$

The n -th EICF convergent is denoted by

$$\frac{p_n^E}{q_n^E} = \langle\langle (b_1, \eta_1), (b_2, \eta_2), \dots, (b_n, \eta_n) \rangle\rangle.$$

By Lemma 3.1, we have the following matrix relation:

$$\begin{pmatrix} b_0 & \eta_0 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} b_1 & \eta_1 \\ 1 & 0 \end{pmatrix} \cdots \begin{pmatrix} b_n & \eta_n \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} q_n^E & \eta_n q_{n-1}^E \\ p_n^E & \eta_n p_{n-1}^E \end{pmatrix}. \quad (30)$$

Since each matrix in (30) belongs to $\Theta \cup \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \Theta$, each EICF convergent p_n^E/q_n^E is an ∞ -rational.

Observe that if $p_n^E(x)/q_n^E(x)$ is of type even/odd, then $1 - p_n^E(x)/q_n^E(x)$ is 1-rational. If $p_n^E(x)/q_n^E(x)$ is of type odd/even, then $1 - p_n^E(x)/q_n^E(x)$ is still of type odd/even.

Proposition 3.8 *Let $x \in (0, 1)$. All rationals of type odd/odd in $\{1 - p_n^E(1-x)/q_n^E(1-x) : n \geq 1\}$ are best 1-rational approximations of x , and hence are OOCF principal convergents of x .*

Proof For each $n \geq 0$, denote by $P_n/Q_n := 1 - p_n^E(1-x)/q_n^E(1-x)$. We have

$$P_n = q_n^E(1-x) - p_n^E(1-x) \text{ and } Q_n = q_n^E(1-x).$$

By the theorem of Short and Walker (see (3); also [23, Theorem 5]), for any $a/b \in \Theta(\infty)$ such that $1 \leq b \leq q_n^E(1-x)$ and $a/b \neq p_n^E(1-x)/q_n^E(1-x)$,

$$|P_n - Q_n x| = |q_n^E(1-x) - p_n^E(1-x) - x \cdot q_n^E(1-x)| < |(1-x)b - a|.$$

For any $c/d \in \Theta(1)$ such that $1 \leq d \leq Q_n$ and $c/d \neq P_n/Q_n$, we have

$$\frac{d-c}{d} = 1 - \frac{c}{d} \in \Theta(\infty) \text{ and } \frac{d-c}{d} \neq p_n^E(1-x)/q_n^E(1-x).$$

Thus $|P_n - Q_n x| < |(1-x)d - (d-c)| = |c - dx|$, which means that P_n/Q_n is a best 1-rational approximation of x . $\square$

The next proposition describes a connection between the principal convergents of OOCF and EICF. Recall that $f(x) = \frac{1-x}{1+x}$ is the conjugacy map defined in Sect. 2.

Proposition 3.9 *Let $x \in (0, 1)$. There is a 1-1 correspondence between the partial quotients of OOCF of x and that of EICF of $f(x)$. In particular, $p_n^E(f(x))/q_n^E(f(x)) = f(p_n(x)/q_n(x))$ for all $n \geq 1$.*

Proof Since

$$f(B(k+1, -1)) = \left[\frac{1}{2k}, \frac{1}{2k-1} \right], \quad f(B(k, 1)) = \left[\frac{1}{2k+1}, \frac{1}{2k} \right],$$

there is a 1-1 correspondence ϕ between the partial quotients of OOCF and EICF as follows:

$$\phi : (k+1, -1) \mapsto (2k, -1), \quad (k, 1) \mapsto (2k, 1).$$

Thus, for $x = \llbracket (a_n, \varepsilon_n)_{n \in \mathbb{N}} \rrbracket$, the EICF expansion of $f(x)$ is $\llbracket \phi(a_n, \varepsilon_n)_{n \in \mathbb{N}} \rrbracket$. Considering the finite expansions, we obtain the second assertion. $\square$

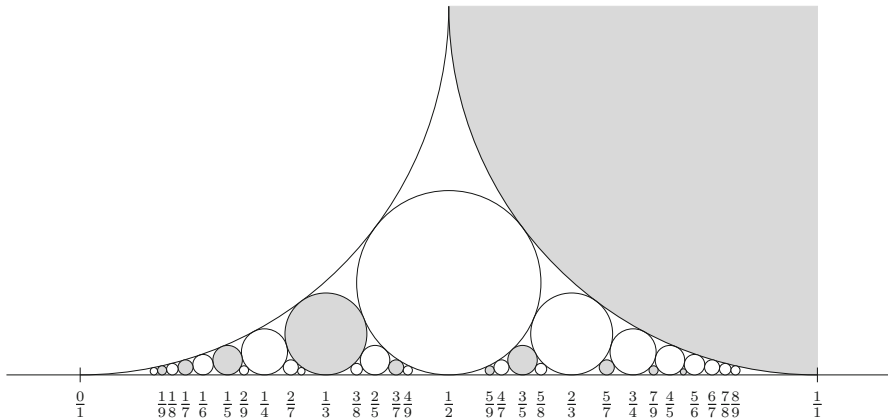


Fig. 2 Ford circles: white circles are based at ∞ -rationals and gray circles are based at 1-rationals (color figure online)

4 Best 1-rational approximation

In this section, we prove Theorem 1.2. Recall that $\mathbb{H}$ is the upper half-plane. The boundary of $\mathbb{H}$ is $\mathbb{R}_\infty = \mathbb{R} \cup \{\infty\}$. Denote by $C_{a/b}$ the horocycle of $\mathbb{H}$ based at a/b whose Euclidean radius is $(2b^2)^{-1}$ and C_∞ the line $\{z = x + i \in \mathbb{H} : x \in \mathbb{R}\}$ (see Fig. 2). We call $C_{a/b}$ a *Ford circle*. The radius of $C_{a/b}$ is denoted by $\text{rad}(C_{a/b})$. We remark that two Ford circles $C_{a/b}$ and $C_{c/d}$ are adjacent to each other if and only if $|ad - bc| = 1$. Let $R_{a/b}(x)$ be the Euclidean radius of the horocycle based at x tangent to $C_{a/b}$. Then by Pythagorean theorem,

$$R_{a/b}(x) = \frac{1}{2}|bx - a|^2.$$

Thus

$$|qx - p| < |bx - a| \iff R_{p/q}(x) < R_{a/b}(x). \quad (31)$$

Since the Ford circles are not overlapped each other, we have

$$\text{rad}(C_{a/b}) \leq R_{c/d}(a/b) \text{ for all } a/b, c/d \in \mathbb{Q}. \quad (32)$$

With these preparations, we are ready to prove our Theorem 1.2.

Proof of Theorem 1.2 Given $x \in \mathbb{R} \setminus \mathbb{Q}$, let us consider its n -th principal convergent p_n/q_n and its n -th pseudo-convergent p_n''/q_n'' . Let $a/b \in \Theta(1)$ such that $a/b \neq p_n/q_n$ and $1 \leq b \leq q_n$. Then

$$\text{rad}(C_{p_n/q_n}) \leq \text{rad}(C_{a/b}). \quad (33)$$

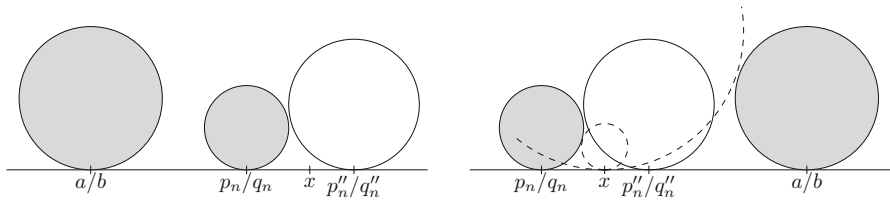


Fig. 3 Two possible relative locations of x , p_n/q_n , a/b and p''_n/q''_n in the proof of Theorem 1.2. The dashed circles are the horocycles based at x tangent to C_{p_n/q_n} and $C_{a/b}$ (color figure online)

By (13) and (14), the Ford circles C_{p_n/q_n} and $C_{p''_n/q''_n}$ are tangent to each other. By Lemma 3.5, x is between p_n/q_n and p''_n/q''_n , then

$$R_{p_n/q_n}(x) \leq \text{rad}(C_{p''_n/q''_n}). \quad (34)$$

Let I_n be the closed interval of endpoints p_n/q_n and p''_n/q''_n . Since $\text{rad}(C_{r/s}) \leq \text{rad}(C_{p_n/q_n})$ for any $r/s \in I_n \cap \mathbb{Q}$, we deduce from (33) (as shown in Fig. 3) that $a/b \notin I_n$. Then we have

$$\text{rad}(C_{p''_n/q''_n}) < R_{a/b}(x). \quad (35)$$

By (34) and (35), we have $R_{p_n/q_n}(x) < R_{a/b}(x)$. Hence by (31), p_n/q_n is a best 1-rational approximation of x .

Conversely, assume that $a/b \in \Theta(1)$ is not a principal convergent of OOCF of x . Then there are consecutive principal convergents p_{n-1}/q_{n-1} and p_n/q_n such that $q_{n-1} \leq b < q_n$ and $a/b \neq p_{n-1}/q_{n-1}$. Thus,

$$\text{rad}(C_{p_n/q_n}) < \text{rad}(C_{a/b}). \quad (36)$$

By (13) and (14), $C_{p_{n-1}/q_{n-1}}$ and C_{p_n/q_n} are tangent to both $C_{p'_n/q'_n}$ and $C_{p''_n/q''_n}$. Without loss of generality, we assume that $p_{n-1}/q_{n-1} < p'_n/q'_n < p''_n/q''_n$ (see Fig. 4). By (36), $a/b \notin [p'_n/q'_n, p''_n/q''_n]$. Now we will show that $R_{p_{n-1}/q_{n-1}}(x) < R_{a/b}(x)$ which by (31), implies that a/b is not a best 1-rational approximation of x . We distinguish three cases.

- (1) If $a/b < p_{n-1}/q_{n-1}$, then obviously $R_{p_{n-1}/q_{n-1}}(x) < R_{a/b}(x)$.
- (2) Now assume $a/b > p''_n/q''_n$. We note that, for $r/s \in \mathbb{Q}$ and $t, t' \in \mathbb{R}$,

$$|t - r/s| < |t' - r/s| \text{ implies that } R_{r/s}(t) < R_{r/s}(t'). \quad (37)$$

Then we have

$$R_{p_{n-1}/q_{n-1}}(x) < R_{p_{n-1}/q_{n-1}}(p''_n/q''_n) = \text{rad}(C_{p''_n/q''_n}) \leq R_{a/b}(p''_n/q''_n) < R_{a/b}(x). \quad (38)$$

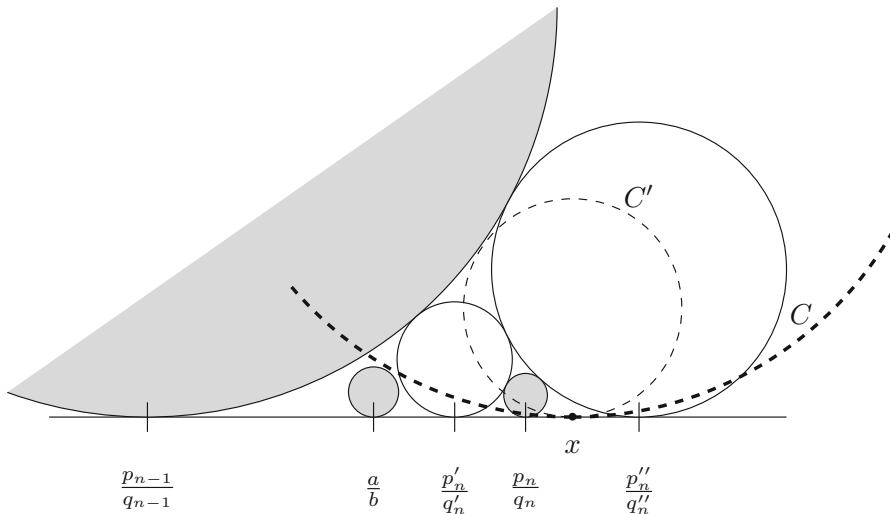


Fig. 4 A possible relative position of x , a/b and the convergents. The dashed circles C and C' are horocycles based at x tangent to $C_{a/b}$ and $C_{p_{n-1}/q_{n-1}}$ (color figure online)

The first and last inequalities in (38) follow from (37) and the fact $p_{n-1}/q_{n-1} < x < p''_n/q''_n < a/b$. The equality in (38) holds since $C_{p''_n/q''_n}$ and $C_{p_{n-1}/q_{n-1}}$ are tangent to each other. The second last inequality in (38) follows from (32).

- (3) Finally, let $a/b \in (p_{n-1}/q_{n-1}, p'_n/q'_n)$. Denote by C and C' the horocycles based at x tangent to $C_{a/b}$ and $C_{p_{n-1}/q_{n-1}}$, respectively (see Fig. 4). Since the tangent point of C and $C_{a/b}$ is an interior point of the area bounded by $C_{p_{n-1}/q_{n-1}}$, $C_{p'_n/q'_n}$ and the real line, we conclude that C intersects $C_{p_{n-1}/q_{n-1}}$. Thus, C is larger than C' , i.e., $R_{p_{n-1}/q_{n-1}}(x) < R_{a/b}(x)$.

□

5 Relation with the regular continued fraction

For simplicity, we denote a RCF as in (1) by $[d_0; d_1, d_2, \dots, d_j, \dots]$. For $1 \leq j \leq d_n$ and for $n \geq 1$, the fractions

$$\frac{p_{n,j}^R}{q_{n,j}^R} = \frac{p_{n-2}^R + jp_{n-1}^R}{q_{n-2}^R + jq_{n-1}^R}$$

are called *the intermediate convergents* (see [12, Section 6] and [17, p.36]). Kraaikamp and Lopes [14] showed that the convergents of EICF are intermediate convergents of RCF. In this section, we show that the OOCF principal convergents are also intermediate convergents of RCF.

The following lemma tells us how the piecewise inverses of OOCF act on RCF expansions.

Lemma 5.1 Let $x = [0; d_1, d_2, \dots]$. Then, the RCF expansion of $f_{(a,\varepsilon)}(x)$ is as follows:

$$f_{(a,\varepsilon)}(x) = \begin{cases} [0; 2, d_1, d_2, \dots] & \text{if } \varepsilon = 1, a = 1, \\ [0; 1, (a-1), 1, d_1, d_2, \dots] & \text{if } \varepsilon = 1, a \geq 2, \\ [0; (d_1+2), d_2, \dots] & \text{if } \varepsilon = -1, a = 2, \\ [0; 1, (a-1), (d_1+1), d_2, \dots] & \text{if } \varepsilon = -1, a \geq 3. \end{cases}$$

Proof If $\varepsilon = 1$, then

$$f_{(a,\varepsilon)}(x) = 1 - [0; a, 1, d_1, d_2, \dots] = \begin{cases} [0; 2, d_1, d_2, \dots] & \text{if } a = 1, \\ [0; 1, a-1, 1, d_1, \dots] & \text{if } a \geq 2. \end{cases}$$

If $\varepsilon = -1$, then

$$\begin{aligned} f_{(a,\varepsilon)}(x) &= 1 - \frac{1}{a - [0; 1, d_1, d_2, \dots]} = 1 - \frac{1}{(a-1) + 1 - [0; 1, d_1, d_2, \dots]} \\ &= 1 - [0; a-1, d_1+1, d_2, \dots] = \begin{cases} [0; d_1+2, d_2, \dots] & \text{if } a = 2, \\ [0; 1, a-1, 1, d_1+1, d_2, \dots] & \text{if } a \geq 3. \end{cases} \end{aligned}$$

□

Applying Lemma 5.1, we have the following theorem.

Theorem 5.2 The OOCF principal convergents of x are intermediate convergents of x .

Proof Let $x = [0; d_1, d_2, \dots] = \llbracket (a_1, \varepsilon_1), (a_2, \varepsilon_2), \dots \rrbracket$. Note that

$$\begin{aligned} x &= f_{(a_1, \varepsilon_1)} \circ f_{(a_2, \varepsilon_2)} \circ \dots \circ f_{(a_k, \varepsilon_k)}(\llbracket (a_{k+1}, \varepsilon_{k+1}), \dots \rrbracket) \quad \text{and} \\ \frac{p_k}{q_k} &= f_{(a_1, \varepsilon_1)} \circ f_{(a_2, \varepsilon_2)} \circ \dots \circ f_{(a_k, \varepsilon_k)}(1). \end{aligned}$$

By Lemma 5.1, x and p_k/q_k have the same prefix in their RCF expansions, except for the last partial quotient of p_k/q_k . Thus, p_k/q_k is an intermediate convergent of x . □

Next, we show that we can convert RCF expansions into OOCF expansions. Before we state the theorem, let us introduce the following notations:

$$\begin{cases} x = [0; d_1, \dots, d_n, \tau] & \text{if } G^n(x) = \tau, \\ x = \llbracket (a_1, \varepsilon_1), (a_2, \varepsilon_2), \dots, (a_n, \varepsilon_n), \gamma \rrbracket & \text{if } T_{\text{OOCF}}^n(x) = \gamma. \end{cases}$$

Theorem 5.3 We can convert RCF expansions into OOCF expansions by the following relations:

$$x = [0; d_1, d_2, \tau] = \begin{cases} \llbracket (2, -1)^{\frac{d_1-1}{2}}, (d_2+1, 1), F(\tau) \rrbracket & \text{if } d_1 \text{ is odd and } \tau \in [\frac{1}{2}, 1), \\ \llbracket (2, -1)^{\frac{d_1-1}{2}}, (d_2+2, -1), F(\tau) \rrbracket & \text{if } d_1 \text{ is odd and } \tau \in [0, \frac{1}{2}), \\ \llbracket (2, -1)^{\frac{d_1}{2}-1}, (1, 1), G(x) \rrbracket & \text{if } d_1 \text{ is even.} \end{cases}$$

Proof Noting $\frac{1}{1+\tau} = 1 - \frac{1}{1+\frac{1}{\tau}}$, we have

$$[0; 1, d_2, \tau] = \frac{1}{1 + \frac{1}{d_2 + \tau}} = 1 - \frac{1}{(d_2 + 1) + \tau}.$$

If $\tau \in [\frac{1}{2}, 1)$, then $\tau = \frac{1}{1+G(\tau)}$. If $\tau \in [0, \frac{1}{2})$, then

$$\tau = 1 - \frac{1}{1 + \frac{1}{[\tau^{-1}] - 1 + G(\tau)}}.$$

Thus $F(\tau) = \frac{1}{[\tau^{-1}] - 1 + G(\tau)}$ if $\tau \in [0, \frac{1}{2})$ and $F(\tau) = G(\tau)$ if $\tau \in (\frac{1}{2}, 1)$. In the case of $d_1 = 1$, we have

$$[0; 1, d_2, \tau] = \begin{cases} \llbracket (d_2 + 1, 1), F(\tau) \rrbracket, & \text{if } \tau \in [\frac{1}{2}, 1), \\ \llbracket (d_2 + 2, -1), F(\tau) \rrbracket, & \text{if } \tau \in [0, \frac{1}{2}). \end{cases} \quad (39)$$

Similarly, if $d_1 = 2$, then

$$x = [0; 2, G(x)] = \frac{1}{2 + G(x)} = 1 - \frac{1}{1 + \frac{1}{1 + G(x)}} = \llbracket (1, 1), G(x) \rrbracket. \quad (40)$$

If $d_1 \geq 3$, i.e., $x \in (0, \frac{1}{3})$, then

$$\begin{aligned} x = [0; d_1, G(x)] &= \frac{1}{d_1 + G(x)} = 1 - \frac{1}{1 + \frac{1}{1 + (d_1 - 2) + G(x)}} \\ &= 1 - \frac{1}{2 - \frac{1}{1 + \frac{1}{(d_1 - 2) + G(x)}}} = \llbracket (2, -1), T_{\text{OOCF}}(x) \rrbracket. \end{aligned} \quad (41)$$

Note that there are $n \in \mathbb{N}$ and $r \in \{0, 1\}$ such that $d_1 - 1 = 2n + r$. Since $T_{\text{OOCF}}(x) = [0; d_1 - 2, G(x)]$, by repeating the process in (41), we have

$$[0; d_1, G(x)] = \llbracket (2, -1)^n, T_{\text{OOCF}}^n(x) \rrbracket \quad \text{and} \quad T_{\text{OOCF}}^n(x) = [0; r + 1, G(x)].$$

Then we can complete the proof by (39) and (40). $\square$

Since p_n^R/q_n^R is a best approximation, if p_n^R/q_n^R is a 1-rational, then p_n^R/q_n^R is a best 1-rational approximation. Thus, by Theorem 1.2, p_n^R/q_n^R is an OOCF convergent. Now we check when an intermediate convergent is an OOCF convergent. Keita [11] proved the following proposition.

Proposition (Keita, Proposition 1.2 in [11]) We have

$$\begin{aligned} q_{n,0}^R = q_{n-2}^R < q_{n-1}^R \leq q_{n,1}^R < \cdots < q_{n,d_n}^R = q_n^R \text{ and} \\ |q_{n,d_n}^R x - p_{n,d_n}^R| = |q_n^R x - p_n^R| < |q_{n-1}^R x - p_{n-1}^R| \\ \leq |q_{n,d_n-1}^R x - p_{n,d_n-1}^R| < \cdots < |q_{n,0}^R x - p_{n,0}^R| = |q_{n-2}^R x - p_{n-2}^R|. \end{aligned}$$

By the above proposition and Theorem 5.2, if p_{n-1}^R/q_{n-1}^R is a 1-rational, then $p_{n,j}^R/q_{n,j}^R$ is not an OOCF principal convergent for any $1 \leq j < d_n$. If p_{n-1}^R/q_{n-1}^R is an ∞ -rational and $p_{n,j}^R/q_{n,j}^R$ is a 1-rational, then $p_{n,j}^R/q_{n,j}^R$ is an OOCF principal convergent.

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References

1. Alperin, R.C.: The modular tree of Pythagoras. *Am. Math. Mon.* **112**(9), 807–816 (2005)
2. Barning, F.J.M.: On Pythagorean and quasi-Pythagorean triangles and a generation process with the help of unimodular matrices. *Math. Cent. Amsterdam Afd. Zuivere Wisk.* **1963**(ZW-011), 37 (1963)
3. Berggren, B.: Pytagoreiska trianglar. *Tidskr. Elementär Mat. Fys. Kemi.* **17**, 129–139 (1934)
4. Boca, F.P., Merriman, C.: Coding of geodesics on some modular surfaces and applications to odd and even continued fractions. *Indag. Math.* **29**(5), 1214–1234 (2018)
5. Cha, B., Kim, D.H.: Intrinsic Diophantine approximation of a unit circle and its Lagrange spectrum (2019) (available at [arXiv:1903.02882](https://arxiv.org/abs/1903.02882))
6. Cha, B., Kim, D.H.: Number theoretical properties of Romik's dynamical system. *Bull. Korean Math. Soc.* **57**(1), 251–274 (2020)
7. Cha, B., Nguyen, E., Tauber, B.: Quadratic forms and their Berggren trees. *J. Number Theory* **185**, 218–256 (2018)
8. Conrad, K.: Pythagorean descent. (2007) (available at <http://www.math.uconn.edu/~kconrad/blurbs/linmultalg/descentPythag.pdf>)
9. Ito, S.: Algorithms with mediant convergence and their metrical theory. *Osaka J. Math.* **26**(3), 557–578 (1989)
10. Jager, H., Liardet, P.: Distributions arithmétiques des dénominateurs de convergents de fractions continues. *Indag. Math.* **91**(2), 181–197 (1988)
11. Keita, A.: Continued fractions and parametric geometry of numbers. *J. Théor. Nr. Bordx.* **29**(1), 129–135 (2017)

12. Khinchin, A.Y.: Continued Fractions, Translated from the 3rd (1961) Russian ed., Reprint of the 1964 Translation. Dover Publications, Mineola (1997)
13. Kraaikamp, C.: A new class of continued fraction expansions. *Acta Arith.* **57**(1), 1–39 (1991)
14. Kraaikamp, C., Lopes, A.: The theta group and the continued fraction expansion with even partial quotients. *Geom. Dedicata.* **59**(3), 293–333 (1996)
15. Moeckel, R.: Geodesics on modular surfaces and continued fractions. *Ergod. Thoery Dyn. Syst.* **2**(1), 69–83 (1982)
16. Panti, G.: Billiards on pythagorean triples and their Minkowski functions. *Discret. Contin. Dyn. Syst.* **40**(7), 4341–4378 (2020)
17. Rockett, A.M., Szűsz, P.: Continued Fractions. World Scientific Publishing Co., Singapore (1992)
18. Romik, D.: The dynamics of Pythagorean triples. *Trans. Am. Math. Soc.* **360**(11), 6045–6064 (2008)
19. Schweiger, F.: Continued fractions with odd and even partial quotients. *Arbeitsber. Math. Inst. Univ. Salzburg.* **4**, 59–70 (1982)
20. Schweiger, F.: On the approximation by continued fractions with odd and even partial quotients. *Arbeitsber. Math. Inst. Univ. Salzburg.* **1**(2), 105–114 (1984)
21. Schweiger, F.: *Ergodic Theory of Fibred Systems and Metric Number Theory*. Oxford University Press, Oxford (1995)
22. Series, C.: The modular surface and continued fractions. *J. Lond. Math. Soc.* **s2–31**(1), 69–80 (1985)
23. Short, I., Walker, M.: Even-integer continued fractions and the Farey tree. In: *Symmetries in Graphs, Maps, and Polytopes Workshop (Springer Proc. in Math. Stat. 159)*, pp. 287–300. Springer, (2016)

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